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Real-time multi-domain collaborative low-light image enhancement for driving safety ⋆

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ABSTRACT

Visual information quality in low-light environments is critical for ensuring autonomous driving safety. Under conditions such as nighttime driving, captured images suffer severe degradation, characterized by color distortion, noise interference, and loss of detail. This not only challenges the performance of visual perception systems but also limits the accuracy of subsequent advanced tasks like object detection, ultimately posing a potential risk to driving safety. To address this challenge, we propose a novel, efficient multi-domain collaborative low-light enhancement network (MCLE-Net) to meet the demands of real-time processing and high-fidelity restoration during low-light driving. The key component is the multi-domain attention block (MDAB), which collaboratively and explicitly processes spatial, channel, and frequency domain image features within a unified architecture. By integrating the fast fourier transform (FFT) into its attention mechanism, it precisely recovers high-frequency details critical for perception tasks while optimizing global contrast. Simultaneously, sparse spatial attention (SSA) and sparse channel attention (SCA) mechanisms adaptively focus on salient features with minimal computational overhead. Furthermore, the network incorporates a U-Net-inspired multi-scale feature extraction block (MSFEB) for progressive feature refinement. A gated fusion block (GFB) dynamically aggregates multi-scale information, improving feature utilization and representational capacity. Extensive experiments on multiple benchmark datasets demonstrate that MCLE-Net achieves an competitive balance between enhancement quality and computational efficiency by significantly improving feature discernibility and detail recovery, providing an efficient and reliable visual enhancement solution for ensuring low-light driving safety.

1. Introduction

The safe and efficient operation of autonomous vehicles in low-light environments heavily relies on the image quality captured by onboard visual sensors, which plays a decisive role in ensuring machine perception. Under insufficient lighting conditions, the effective detection range of vehicle cameras is significantly reduced. The captured images typically suffer from issues such as contrast degradation, color distortion, loss of detail, and noise interference, resulting in a sharp decline in clarity (Lin et al., 2025). This severe degradation of image quality adversely impacts subsequent advanced visual tasks (such as environmental perception, object detection, and semantic segmentation), thereby jeopardizing driving safety (Tao et al., 2024; Wang et al., 2022a). Consequently, developing efficient forward-looking low-light image enhancement (LLIE) technology is of paramount importance for ensuring the safe and reliable operation of driving systems.

Currently, strategies for improving image quality in low-light environments primarily fall into two categories: hardware solutions and software algorithms. Hardware approaches enhance imaging quality by increasing the sensitivity of camera sensors, or optimizing the collaboration between sensors and processing systems. However, these methods usually encounter problems such as high costs, complex integration, and strict requirements for performance and reliability. In addition, the inherent noise of the device introduced by factors such as dark current and electronic noise also increases the complexity of hardware design (Shi et al., 2024).

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In contrast, software-based LLIE algorithms offer a more flexible and efficient pathway and have become a research hotspot. Traditional LLIE algorithms mostly rely on physical models or manually designed prior knowledge (Ancuti & Ancuti, 2013; Berman et al., 2016), with the most representative approaches including methods based on histogram equalization and Retinex theory. Classic Retinex algorithms such as single-scale Retinex (Shen et al., 2017) and multi-scale Retinex (Jobson et al., 1997a) and their subsequent improvements (Zhao & Shi, 2026) have improved image brightness to a certain extent. However, these traditional approaches often struggle to avoid inherent flaws such as local over-enhancement, color distortion, or detail loss, making them unsuitable for demanding real-time applications such as autonomous driving.

In recent years, deep learning-based LLIE methods have become mainstream. These studies no longer rely on manually designed priors, but instead use deep neural networks, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), or generative adversarial networks (GANs), to directly learn complex nonlinear mappings from data (He et al., 2025; Li et al., 2026; Liang et al., 2025; Lin et al., 2024; Yin & Liu, 2025; Zhang et al., 2023b), thereby surpassing traditional methods in preserving image details and color fidelity (Lu et al., 2025). To capture long-range dependencies in images, researchers have also introduced Transformer models. For example, Cai et al. (2023a) combined Retinex theory with Transformer models, achieving significant improvements in enhancement effects. Subsequently, diffusion models have gained significant attention (Jiang et al., 2024a). By integrating pre-trained models such as Stable Diffusion (Rombach et al., 2022) and ControlNet (Zhang et al., 2023a), they can generate high-quality results rich in detail. However, their inherent iterative sampling process incurs high computational costs and slow inference speeds, severely limiting their deployment in practical applications.

Although existing LLIE methods significantly improve image visual quality, their performance remains insufficient for demanding downstream visual tasks in safety-critical applications. The core challenge lies in the difficulty of simultaneously balancing visual quality, model inference efficiency, and performance gains for downstream tasks. To address these challenges, we propose an efficient multi-domain collaborative low-light enhancement network (MCLE-Net). By collaboratively perceiving local spatial structures, global channel semantics, and frequency domain energy distributions, this framework comprehensively models image content and degradation patterns with minimal computational overhead. It achieves a highly competitive trade-off among visual quality, inference speed, and downstream task performance. As shown in Fig. 1, MCLE-Net outperforms existing mainstream methods in quantization metrics, inference speed, and visual quality, showcasing its potential for real-time applications.

Specifically, MCLE-Net overall architecture consists of cascaded feature extraction module (FEM) integrated with residual connections, ensuring the preservation of low-frequency information in the original image. This accelerates model convergence and stabilizes the training process. Each FEM incorporates two tandem U-Net-inspired multi-scale feature extraction blocks (MSFEBs) that are used for progressive feature optimization, aiming to address various low-light degradation issues such as from global brightness loss to suppression of fine details. The core innovation of our network is the multi-domain attention block (MDAB), which synergistically processes information across the spatial, channel, and frequency domains. Through sparse spatial attention (SSA) and sparse channel attention (SCA), the MDAB selectively restores critical structures and recalibrates feature representations. Furthermore, it innovatively integrates the fast fourier transform (FFT), directly manipulating frequency components for precise recovery of high-frequency details. Finally, the gated fusion block (GFB) serves as a dynamic gating mechanism to adaptively merge feature streams, ensuring optimal information flow and utilization throughout the network.

Our main contributions are summarized as follows:

1) We propose the MCLE-Net, which achieves collaborative modeling of spatial, channel, and frequency domains information within a unified multi-scale learning framework. Through a gated fusion mechanism, it dynamically aggregates multi-source information to efficiently reconstruct degraded images and critical details in low-light environments.
2) We design the core MDAB, innovatively employing FFT for high-frequency detail recovery and combining sparse attention to adaptively focus on critical information with minimal computational overhead.
3) We conduct comprehensive experimental evaluations on datasets including LoLI-Street, LOL v1, LOL v2, and LSRW. Extensive quantitative and qualitative comparisons demonstrate that MCLE-Net achieves competitive performance on key metrics. Furthermore, ablation studies validate the effectiveness of core modules and their synergistic interactions.

This paper is organized as follows. Section 2 reviews related work. Section 3 details the architecture of our proposed MCLE-Net. Section 4 validates and compares its performance through comprehensive experiments. Section 5 concludes the paper and outlines future work.

2. Related work

2.1. Traditional LLIE methods

Traditional LLIE methods primarily include histogram equalization (Abdullah-Al-Wadud et al., 2007; Arici et al., 2009; Celik & Tjahjadi, 2011; Ibrahim & Kong, 2007; Lee et al., 2013), gamma correction, and Retinex theory.

Histogram equalization optimizes image contrast and brightness by adjusting the distribution of pixel intensities (Arici et al., 2009). Celik and Tjahjadi (2011) designed a universal image contrast enhancement framework based on histogram equalization, formulating it as an optimization problem that minimizes a cost function. Gamma correction enhances brightness and contrast through nonlinear transformation of image grayscale values. For instance, Huang et al. (2012) combined gamma correction with histogram equalization to address the issue of non-uniform enhancement caused by fixed parameters. Methods based on Retinex theory (Fu et al., 2016) posit that human color perception results from the combined effects of reflected light and ambient illumination. By decomposing images into reflectance and illumination layers, processing them separately, and then reconstructing the image, these approaches achieve enhancement (Jobson et al., 1997b). For example, Guo et al. (2016) proposed an LLIE method based on illuminance map estimation, decomposing images into the product of reflectance and illumination components, enabling enhancement by estimating a single illuminance map. Similarly, Li et al. (2018) designed a structure-revealing LLIE method based on a robust Retinex model.

While these traditional LLIE methods have achieved certain results in specific scenarios, they generally suffer from inherent limitations. For instance, global adjustment methods such as histogram equalization and gamma correction are prone to artifacts and color distortion. Methods based on Retinex theory heavily rely on cumbersome manual parameter tuning and prior assumptions, and carry the risk of amplifying noise during processing.

2.2. LLIE method based on deep learning

In recent years, deep learning-based LLIE methods have become mainstream, primarily categorized into supervised and unsupervised strategies.

Supervised learning methods directly learn an end-to-end mapping between low-light and normal-light images by training on large datasets

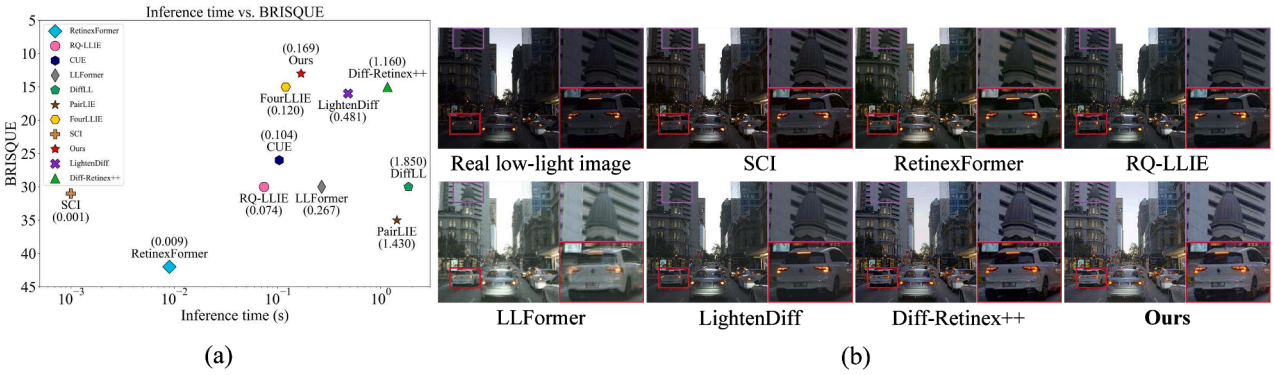

Fig. 1. Comparison of the proposed MCLE-Net and other models on real low-light images. (a) Quantitative comparison on BRISQUE (↓) and inference time (↓). (b) Qualitative comparison results.

of paired images. Early works, such as LLNet (Lore et al., 2017) and studies inspired by Retinex theory (Hu et al., 2021; Huang et al., 2022; Lv et al., 2021; Wu et al., 2023), decomposed images into illumination and reflectance components for separate enhancement. While achieving preliminary success, these approaches often suffered from artifacts or detail loss. To address specific degradation issues, subsequent research designed more targeted network architectures. For instance, NDAT-Net (Liu et al., 2021) introduced specialized modules to decouple contrast enhancement from denoising. Meanwhile, reference (Jiang et al., 2024c) focused on mitigating loss of information and noise interference under extreme conditions, through dedicated exposure difference and global color calibration modules. Recently, Transformer architectures have demonstrated significant potential. For instance, UPT-Flow (Xu et al., 2025) modifies Transformer attention calculations by incorporating prior knowledge to reduce information distortion. However, its performance is often constrained by the generalization capabilities of the relied-upon prior knowledge.

To overcome reliance on paired data, unsupervised methods have emerged (He et al., 2025). These approaches typically leverage architectures like GANs to learn mappings from low-light to normal-light domains on unpaired data. Within this paradigm, various strategies have been proposed. For instance, Xing et al. (2023) introduced CLEGAN, which employs contrastive learning to preserve content consistency. Dai et al. (2024) proposed the ImCam framework by combining the Retinex model with the generative adversarial network. Wang et al. (2022b) proposed MAGAN, guided by hybrid attention, to model complex relationships between pixels and features. Subsequently, diffusion models have also been applied, such as LightDiff (Li et al., 2024) and LightenDiffusion (Jiang et al., 2024b), which employ techniques like multi-conditional control or Retinex decomposition within the latent space to further improve enhancement quality. However, unsupervised methods, lacking strong pixel-level constraints, often struggle to match the visual fidelity of leading supervised approaches in precisely restoring fine textures and suppressing artifacts and noise.

While the aforementioned methods improve visibility in low-light scenarios, an increasing number of studies are emerging that serve downstream advanced vision tasks, such as object detection. Recent work has explored strategies to enhance perception performance, including diffusion models (Li et al., 2024), efficient "one-stop" frameworks (Li et al., 2025), multimodal fusion (Xian et al., 2024), and data augmentation (Zheng et al., 2023b). However, these approaches (Huang et al., 2024; Liu et al., 2024; Xue et al., 2024) share a common limitation: their network designs primarily rely on processing in the spatial or channel domain, lacking explicit modeling and precise manipulation of the image frequency components. This results in limited capabilities for recovering edge and texture details crucial for downstream tasks. To address this limitation, frequency-aware LLIE methods have been proposed to separate illumination distributions from structural textures. Early frequency-aware approaches, such as FourLLIE (Wang et al., 2023a), leveraged fourier frequency information to guide the enhancement process. Recent works have further advanced this approach. For instance, SWANet (He et al., 2024b) incorporated wavelet-based attention mechanisms specifically tailored to perceive and suppress local noise amplified during illumination adjustment. Similarly, WalMaFa (Tan et al., 2025) utilized a wavelet-based Mamba block to enhance global brightness and applied fourier adjustments to refine local textures. Furthermore, generative approaches like FWDM (He et al., 2024a) and WZD (Peng et al., 2025) embedded fourier or wavelet priors into diffusion models to constrain the generative sampling process. Other approaches, such as Lu-former (Lu et al., 2025) and OFNet (Kong et al., 2025), introduced frequency augmentation or frequency-domain loss functions to explicitly regularize the network and assist in capturing rich global dependencies.

Despite these advancements, existing frequency-aware models face two main limitations. First, holistic frequency manipulation, such as applying convolutions across the entire spectrum, inherently alters the phase, risking structural distortion and edge artifacts. Second, isolated processing or heavy sequential cascading hinders efficient, real-time synergy between global frequency and local spatial semantics. To address these shortcomings, MCLE-Net introduces a new strategy. Unlike prior works, our MDAB ensures structural preservation via phase-preserving decoupling, applying learnable filters solely to the amplitude spectrum while leaving the phase completely unmodified. Furthermore, MCLE-Net establishes explicit multi-domain synergy by dynamically fusing this frequency recovery with computationally efficient sparse spatial and channel attention, achieving high-fidelity restoration.

3. Proposed method

In this section, we introduce the proposed MCLE-Net and the loss function used in detail.

3.1. Overview of MCLE-Net

The overall architecture of our proposed MCLE-Net is shown in Fig. 2(a). Given a low-light image $I' \in \mathbb{R}^{H\times W\times 3}$, the network first extracts shallow features through a 3×3 convolutional layer, obtaining the feature map $X_0 \in \mathbb{R}^{H\times W\times C}$, where C represents the intermediate feature dimension.

$$X_0 = Conv_{3\times 3}(I') \tag{1}$$

Subsequently, X_0 is fed into N cascaded core functional modules FEM for iterative extraction of deep features. Through this cascaded design, the network progressively restores image quality, focusing on coarse-grained adjustments in the early stages and performing fine detail enhancements in the later stages. Each FEM module consists of two serial MSFEBs and a 3×3 convolutional layer, incorporating a local residual

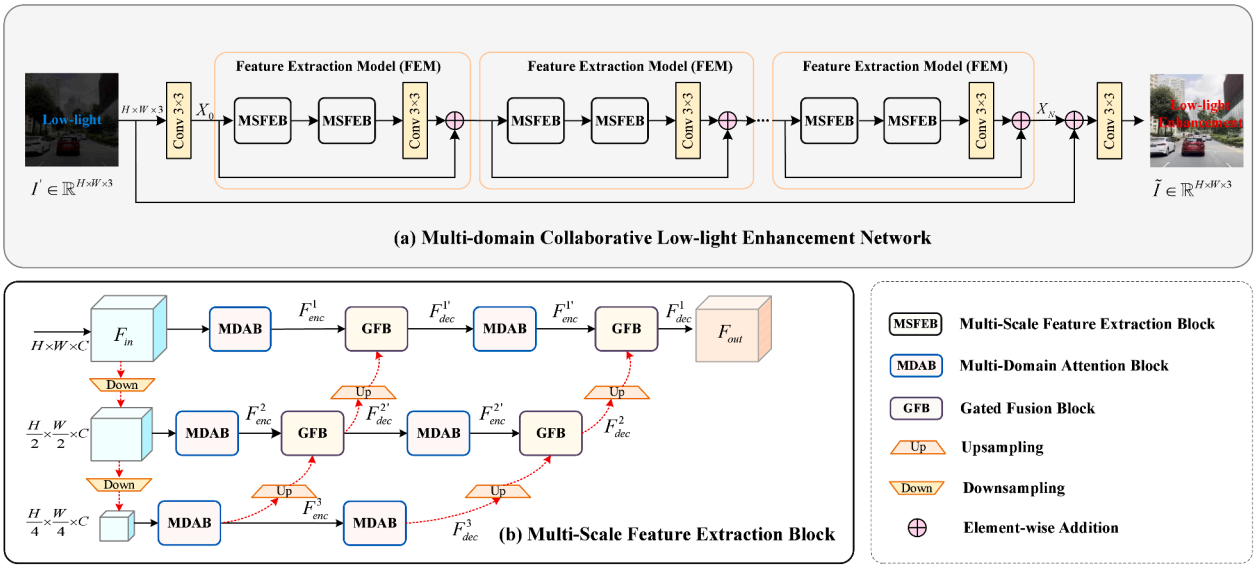

Fig. 2. The network architecture of our proposed method. (a) Multi-domain collaborative low-light enhancement network (MCLE-Net). (b)Multi-scale feature extraction block (MSFEB).

connection. Specifically, the ith FEM module receives X_{i-1} as input, and its processing can be represented as

$$\begin{cases} Y_1 = \mathcal{H}_{MSFEB}(X_{i-1}) \\ Y_2 = \mathcal{H}_{MSFEB}(Y_1) \\ X_i = Conv_{3\times3}(Y_2) \oplus X_{i-1}, i = 1, \dots, N \end{cases} \tag{2}$$

where $\oplus$ represents element-wise addition, $\mathcal{H}_{MSFEB}$ represents the MSFEB function, whose detailed definition will be elaborated in subsequent subsections. Finally, to preserve the low frequency information of the original image and accelerate convergence, the output X_N from the last FEM module is added to the initial shallow feature $I^{'}$ via global residual connection. The fused features then pass through a 3×3 convolutional layer to yield the final enhanced image $\widetilde{I} \in \mathbb{R}^{H\times W\times 3}$:

$$\widetilde{I} = Conv_{3\times3}(I^{'} \oplus X_N) \tag{3}$$

3.2. Multi-scale feature extraction block

Low-light image degradation is an inherent multiscale problem, which manifests in two aspects. On one hand, the degradation itself exhibits multiscale properties, including large-area brightness loss (macroscale) and minute noise and artifacts (microscale). On the other hand, scene information of image content is inherently distributed across different scales, for instance, macro-level scene structure and micro-level object texture. Consequently, single-scale processing methods struggle to simultaneously account for both global context and local details.

To effectively address this challenge, we designed the MSFEB, whose detailed structure is illustrated in Fig. 2(b). This module functions as a dynamic system that collaboratively optimizes multi-level feature representations through iterative cross-scale interactions, employing a three-tiered symmetrical encoder-decoder architecture.

3.2.1. Encoding path

The input feature F_{in} first enters the main trunk (Level 1). Subsequently, through two cascaded downsampling operations, input features for levels 2 and 3 are generated, thereby constructing a hierarchical feature pyramid while simultaneously reducing spatial dimensions and expanding the receptive field. At the initial stage of each level i ($i \in \{1, 2, 3\}$), we deploy the core feature extraction unit MDAB to capture rich contextual information at the current scale. Its output F^i_{enc} can be uniformly represented as

$$F^i_{enc} = E_i\left(Down_i\left(F_{in}\right)\right) \tag{4}$$

where $Down_i(\cdot)$ denotes the ith downsampling operation performed on F_{in} (when $i = 1$, this operation is an identity mapping). $E_i(\cdot)$ represents the execution of the MDAB operation on level i.

3.2.2. Decoding path

This process iterates in a bottom-up manner, integrating high-level abstract semantic information into low-level fine spatial details. Spatial resolution is progressively restored during this path through upsampling operations. Specifically, features from the deepest layer (Level 3, the bottleneck layer) serve as the starting point for information fusion after undergoing MDAB deep refinement stages. For any level i ($i \in \{1, 2\}$) along the decoder path, its output feature F^i_{dec} can be expressed as

$$F^i_{dec} = D_i\left(Up\left(F^{i+1}_{dec}\right), F^i_{enc}\right) \tag{5}$$

where $Up(\cdot)$ denotes the upsampling operation. $D_i(\cdot)$ fuses features F^{i+1}_{dec} from deeper layers and features F^i_{enc} from the same-level encoder via GFB (when $i = 2$, $F^{i+1}_{dec} = F^{i+1}_{enc}$).

Through this bottom-up, fusion then refinement iterative mechanism, it effectively integrates deep semantic information with shallow spatial details, ultimately outputting highly refined features $F_{out} = F^1_{dec}$ at the original resolution (level 1).

3.3. Multi-domain attention block

To address the limitations of conventional methods that operate solely within the spatial or channel domain, we introduce the MDAB, a module designed to perform collaborative feature processing across three domains: spatial, channel, and frequency. MDAB serves as the core unit for feature extraction and refinement within our network. The detailed structure of MDAB is illustrated in Fig. 3(a). To efficiently capture feature dependencies across different dimensions, MDAB employs a divide-and-conquer strategy. It decouples the input features into two parallel branches: the spatial channel domain and the frequency domain, each equipped with a dedicated attention mechanism.

Specifically, for the input feature map $F_0 \in \mathbb{R}^{H\times W\times C}$, it is first uniformly split along the channel dimension into two parallel sub-features $F_1, F_2 \in \mathbb{R}^{H\times W\times \frac{C}{2}}$:

$$\left[F_1, F_2\right] = Split(F_0) \tag{6}$$

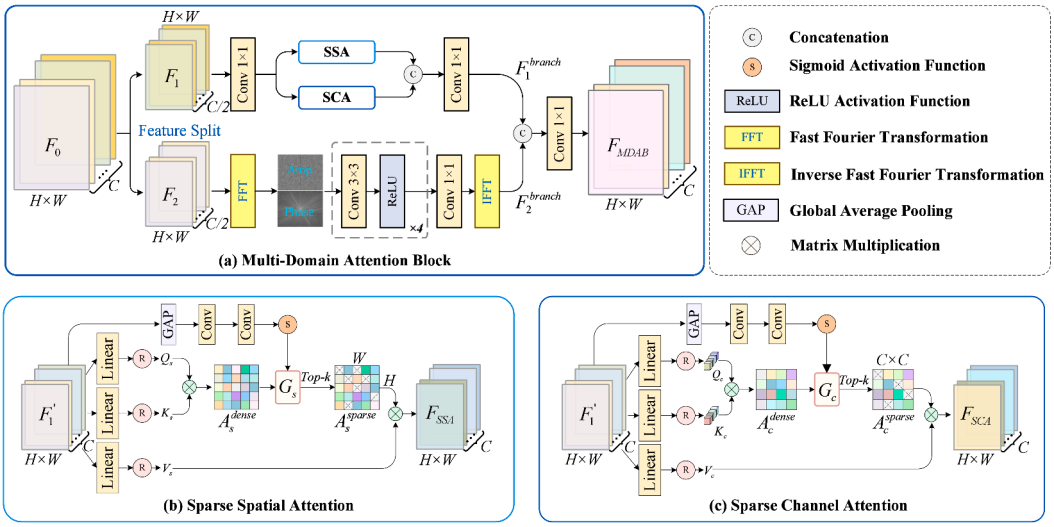

Fig. 3. The network architecture of essential modules. (a) Multi-Domain Attention Block (MDAB). (b) Sparse Spatial Attention (SSA). (c) Sparse Channel Attention (SCA).

Subsequently, F_1 and F_2 are processed by the spatial channel domain path and the frequency domain path, respectively. Finally, the outputs from both branches are concatenated and fused by a 1×1 convolutional layer for cross-domain information fusion, achieving synergistic enhancement. The overall process can be summarized as follows.

$$F_{MDAB} = Conv_{1\times1}(Concat(\mathcal{H}_{sc}(F_1), \mathcal{H}_{freq}(F_2))) \tag{7}$$

where $\mathcal{H}_{sc}(\cdot)$ and $\mathcal{H}_{freq}(\cdot)$ are the processing functions for the spatial channel domain and the frequency domain, respectively, as detailed below.

3.3.1. Spatial and channel domain branch

This path $\mathcal{H}_{sc}(\cdot)$ processes the feature map F_1, with the primary objective of explicitly modeling the local spatial structural dependencies and inter-channel relationships of the image. First, F_1 undergoes an initial feature transformation by a 1×1 convolution to yield $F_1^{'}$. Subsequently, $F_1^{'}$ is fed in parallel into the SSA and SCA modules.

Sparse spatial attention: The SSA module is designed to efficiently model long-range dependencies between arbitrary spatial locations in the feature map. As shown in Fig. 3(b), it consists of a self-attention path and a parallel spatial gating pathway, which operate collaboratively to perform a sparsified attention computation.

For the input feature $F_1^{'}$, the self-attention path generates Query (Q), Key (K), and Value (V) matrices through three parallel linear transformation layers. These matrices are then reshaped to $Q_s, K_s, V_s \in \mathbb{R}^{HW\times C}$. Subsequently, by calculating the matrix multiplication of Q_s and K_s , applying the Softmax function, obtaining a dense global spatial attention map $A_s^{dense} \in \mathbb{R}^{HW\times HW}$:

$$A_s^{dense} = Soft\max\left(\frac{Q_s K_s^T}{\sqrt{d_k}}\right) \tag{8}$$

where d_k represents the dimension of the Key vector. Each element (i, j) of A_s^{dense} represents the attention weight of spatial position i to position j.

Meanwhile, the parallel spatial gating path evaluates the importance of each spatial location and generates a spatial importance mask. Using global average pooling (GAP) aggregates the global spatial information of $F_1^{'}$, and which is then passed through two convolutional layers and Sigmoid function to produce the normalized spatial gating map $G_s \in \mathbb{R}^{H\times W\times1}$:

$$G_s = Sigmoid(Conv_{1\times1}(Conv_{1\times1}(GAP_c(F_1^{'})))) \tag{9}$$

To sparsify the attention map and reduce computational complexity, we use G_s (reshaped to $\mathbb{R}^{HW\times1}$) to weightedly modulate the dense attention map A_s^{dense}. An adaptive Top-k function is then applied to each row of the modulated attention map, utilizing a threshold filtering mechanism. Specifically, it retains only the attention weights greater than a predefined threshold τ (set to 0.6), while masking the rest to zero. In this formulation, the equivalent k becomes highly adaptive to the input image content and feature correlations. This sparsity constraint is critical for computational efficiency and forces the model to focus only on the most salient spatial relationships. The sparse attention map A_s^{sparse} is obtained as:

$$A_s^{sparse} = adaptive_Top\text{-}k(A_s^{dense} \odot G_s) \tag{10}$$

where $\odot$ denotes element-wise broadcast multiplication. Finally, A_s^{sparse} is multiplied with the value matrix V_s to aggregate feature information from key spatial locations, and the result is reshaped to obtain the output F_{SSA} of the SSA module:

$$F_{SSA} = Reshape(A_s^{sparse} \otimes V_s) \tag{11}$$

Sparse channel attention: The SCA module is structurally symmetric to SSA, but all its operations are conducted along the channel dimension. It is designed to model and recalibrate inter-channel dependencies. For the input feature $F_1^{'}$, SCA also employs a collaborative self-attention and gating mechanism. It generates channel-dimensional Query, Key, Value matrix $Q_c, K_c, V_c \in \mathbb{R}^{C\times HW}$, and computes the dense channel attention map $A_c^{dense} \in \mathbb{R}^{C\times C}$:

$$A_c^{dense} = Soft\max\left(\frac{Q_c K_c^T}{\sqrt{d_k}}\right) \tag{12}$$

Analogous to SSA, the parallel channel gating path processes the input feature $F_1^{'}$ using GAP and convolutional operations to generate a channel gating map $G_c \in \mathbb{R}^{C\times1\times1}$. Subsequently, G_c and the adaptive $Top\text{-}k$ function are used to generate and sparsify the channel attention map A_c^{sparse}, which is then multiplied by the value matrix V_c to obtain the output F_{SCA}:

$$F_{SCA} = Reshape(adaptive_Top\text{-}k(A_c^{dense} \odot G_c) \otimes V_c) \tag{13}$$

Finally, F_{SSA} and F_{SCA} are concatenated and fused via a 1×1 convolution to produce the final output feature map F_1^{branch} of this branch.

3.3.2. Frequency domain branch

This branch $\mathcal{H}_{freq}(\cdot)$ provides a global perspective for low-light enhancement that is complementary to the spatial domain. While spatial

domain operations excel at capturing local details, frequency domain analysis is more adept at capturing the overall structure and periodic patterns of image, such as contrast, brightness, and periodic noise. Physically, the amplitude spectrum of an image determines the energy of its frequency components, which directly governs its contrast, brightness, and the sharpness of edges and textures. The phase spectrum, in contrast, encodes the spatial location of these components, thus defining the core structure and contours of objects. By operating in this domain, we can directly manipulate these fundamental properties.

First, the input spatial feature map $F_2 \in \mathbb{R}^{H\times W\times \frac{C}{2}}$ is transformed into the frequency domain via a 2D discrete fourier transform (DFT), which is implemented efficiently as an FFT. This decomposes the feature map into its amplitude spectrum ($\mathcal{A}$) and phase spectrum ($\mathcal{P}$):

$$\mathcal{F}(u,v) = \sum_{x=0}^{H-1}\sum_{y=0}^{W-1} F_2(x,y)e^{-j2\pi(\frac{ux}{H}+\frac{vy}{W})} = \mathcal{A}(u,v)e^{j\mathcal{P}(u,v)} \tag{14}$$

where u and v represent the horizontal and vertical frequencies, respectively. x and y represent the row and column coordinate in the feature map. The amplitude spectrum $\mathcal{A}$ governs the contrast and noise level of the image, whereas the phase spectrum $\mathcal{P}$ encodes the core structural and positional information of objects.

Then the amplitude spectrum $\mathcal{A}$ is passed through sub-network $\mathcal{G}_{amp}$ composed of convolutional layers and ReLU activation functions. This network acts as a learnable, adaptive frequency filter, capable of correcting for low-light degradation patterns. The correction process can be expressed as a convolution in the frequency domain:

$$\mathcal{A}' = \mathcal{G}_{amp}(\mathcal{A}) \tag{15}$$

where $\mathcal{G}_{amp}$ represents the convolutional sub-network for amplitude spectrum correction, and $\mathcal{A}'$ is the corrected amplitude spectrum.

Meanwhile, the phase spectrum $\mathcal{P}$, which contains the core structural information of the image, is preserved without modification. Since the underlying spatial layout is physically invariant to illumination changes, fixing the phase acts as a structural prior. This design choice ensures that object contours and structures are not geometrically distorted while adjusting global visual properties, thereby preventing common enhancement artifacts such as ghosting. Furthermore, while the frequency branch safely regulates global illumination and contrast via amplitude modulation, the parallel spatial channel domain branch is explicitly responsible for local structural refinement and noise suppression. This decoupled strategy ensures robust illumination enhancement without risking structural integrity.

Finally, $\mathcal{A}'$ is recombined with the original phase spectrum $\mathcal{P}$, and transformed back to the spatial domain via the inverse fast fourier transform $iFFT(\cdot)$ to generate the output feature map F_2^{branch} of this branch:

$$F_2^{branch} = \mathcal{H}_{freq}(F_2) = iFFT(\mathcal{A}' \cdot e^{j\mathcal{P}}) \tag{16}$$

Through this process, F_2^{branch} is endowed with enhanced visual properties.

3.4. Gated fusion block

To effectively integrate the shallow features X_1 from the encoder and the deep features X_2 (after upsampling) from the decoder within the MSFEB, we have designed the GFB. The detailed structure of the GFB is illustrated in Fig. 4.

Conventional feature fusion methods, such as simple concatenation or element-wise addition, are unable to dynamically adjust the weights of the information flow based on the content of the input features. This may lead to the dilution of important features or the introduction of irrelevant noise. To overcome this limitation, GFB employs a data-driven gating mechanism. By analyzing the content of the two input paths in real-time, it dynamically and adaptively generates fusion weights, thereby enabling selective information aggregation.

First, for the two input feature maps $X_1, X_2 \in \mathbb{R}^{H\times W\times C}$, we process them through a subnetwork consisting of convolutional layers and a GAP operation, to extract their respective global information description vectors $v_1, v_2 \in \mathbb{R}^{1\times 1\times C}$:

$$\begin{cases} v_1 = GAP(Conv_{1\times1}(Conv_{3\times3}(X_1))) \\ v_2 = GAP(Conv_{1\times1}(Conv_{3\times3}(X_2))) \end{cases} \tag{17}$$

Subsequently, we concatenate v_1 and v_2, then process the result through a gating network, consisting of a 1×1 convolutional layer, a fully connected (FC) layer, and a Sigmoid function, to generate a unified gating vector W_{gate}:

$$W_{gate} = Sigmoid(FC(Conv_{1\times1}(Concat(v_1, v_2)))) \tag{18}$$

Using channel split operation, the W_{gate} is uniformly divided along the channel dimension to obtain two independent weight vectors, w_1 and w_2. These vectors are used to dynamically modulate the input features X_1 and X_2, respectively, through channel-wise weighting. Finally, the weighted features are concatenated and fused to produce the final output feature map Y:

$$Y = Conv_{1\times1}(Concat(w_1 \otimes X_1, w_2 \otimes X_2)) \tag{19}$$

3.5. Loss function

To effectively supervise the training of the MCLE-Net, we use a hybrid loss function. It is composed of three weighted components: $\mathcal{L}_{L_1}$ loss, frequency domain loss $\mathcal{L}_{freq}$, and multi-scale structural similarity (MS-SSIM) loss $\mathcal{L}_{MS-SSIM}$. The total loss $\mathcal{L}_{total}$ is expressed as

$$\mathcal{L}_{total} = \lambda_1 \cdot \mathcal{L}_{L_1} + \lambda_2 \cdot \mathcal{L}_{freq} + \lambda_3 \cdot \mathcal{L}_{MS-SSIM} \tag{20}$$

where λ_1, λ_2, and λ_3 are weight hyperparameters that balance the contributions of each term. We set the weights to $\lambda_1 = 1.0$, $\lambda_2 = 0.1$, and $\lambda_3 = 0.01$.

3.5.1. $\mathcal{L}_{L_1}$ loss

The $\mathcal{L}_{L_1}$ loss is used to ensure the pixel-level accuracy of generated images, which is given by

$$\mathcal{L}_{L_1} = \frac{1}{N}\sum_{i=1}^{N}\left|I_{out} - I_{gt}\right| \tag{21}$$

where N is the total number of pixels. I_{out} represents the output image of MCLE-Net, and I_{gt} is the corresponding high-quality ground truth (GT) image.

3.5.2. Frequency domain loss

To effectively supervise the frequency processing branch of the network, we designed a frequency domain loss. It is defined by calculating the L_1 distance between the FFT results of the enhanced image and the GT image, which is given by

$$\mathcal{L}_{freq} = \left\|\mathcal{F}(I_{out}) - \mathcal{F}(I_{gt})\right\|_1 \tag{22}$$

where $\mathcal{F}(\cdot)$ denotes the FFT operation and $\|\cdot\|_1$ represents the L_1 norm. Specifically, this L_1 distance is computed directly over both the real and imaginary components to jointly constrain the amplitude and structural phase, without additional frequency weighting or normalization. By evaluating the error directly in the frequency domain as a complementary constraint, the discrepancy at every frequency bin is explicitly exposed and penalized. This formulation prevents the network from ignoring high-frequency components to satisfy global brightness shifts, thereby implicitly compelling the model to preserve high-frequency details, such as edges and textures, and restore a natural spectral distribution.

3.5.3. MS-SSIM loss

To better align with human visual perception, we incorporate the MS-SSIM loss. It calculates structural similarity across multiple scales, based on the SSIM (Wang et al., 2004) metric, offering a more robust

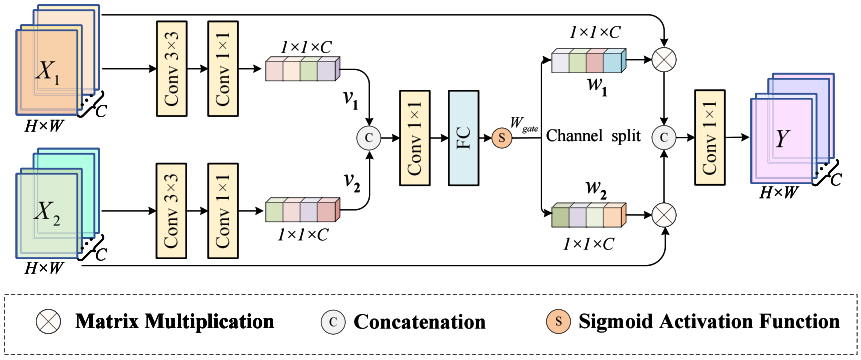

Fig. 4. The network architecture of gated fusion block (GFB).

Table 1
The specific details of the datasets used in our experiments.

Datasets	Source	Type	Training images	Test images
LoLI-Street	Islam et al. [ACCV], 2024	real + synthetic	33,000	1000
LOL v1	Wei et al. [BMVC], 2018	real	485	15
LOL v2-real	Yang et al. [CVPR], 2020	real	689	100
LOL v2-synthetic	Yang et al. [CVPR], 2020	synthetic	900	100
LSRW	Hai et al. [JVCIR], 2023	real	5600	50

quality evaluation than a single-scale approach. Its formulation combines luminance (l), contrast (c), and structure (s) comparison terms:

$$\mathcal{L}_{MS-SSIM} = 1 - \left[l_M(I_{\text{out}}, I_{\text{gt}})\right]^{\alpha_M} \cdot \prod_{j=1}^{M} \left[c_j(I_{\text{out}}, I_{\text{gt}})\right]^{\beta_j} \left[s_j(I_{\text{out}}, I_{\text{gt}})\right]^{\gamma_j} \tag{23}$$

where M is the number of scales, and α_M, β_j, γ_j are parameters that weight the relative importance of the luminance, contrast, and structure components across different scales. Further details regarding the formulation of MS-SSIM are available in Wang et al. (2003).

4. Experiments and discussion

In this section, we present a comprehensive evaluation of our proposed MCLE-Net. We first detail the experimental setup, including the datasets, evaluation metrics, and comparison methods. Subsequently, we provide extensive quantitative and qualitative comparisons against existing methods on multiple benchmark datasets. Furthermore, we conduct ablation studies to validate the effectiveness of the key components within our model. Finally, an analysis of model complexity and inference speed is presented to assess its potential for real-world applications.

4.1. Experimental setup

4.1.1. Datasets

To ensure the comprehensiveness and robustness of evaluation, our experiments are conducted on several LLIE benchmarks datasets with diverse characteristics. Our primary training and evaluation are performed on the LoLI-Street dataset (Islam et al., 2024), a large-scale dataset for autonomous driving scenes, which is categorized into three image degradation levels: dense, moderate, and light. To further validate the generalization capabilities of our model, we also conduct evaluations on several other widely used datasets, including LOL v1 (Wei et al., 2018), LOL v2 (real and synthetic) (Yang et al., 2020), and the large-scale surveillance dataset LSRW (Hai et al., 2023). The specific details of the datasets used are shown in Table 1.

4.1.2. Evaluation metrics

We assess the performance of all methods from three distinct perspectives. **Full-Reference Fidelity Metrics:** Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), SSIM, MS-SSIM, and Mean Absolute Error (MAE) to measure fidelity to the GT reference image. **Perceptual metrics:** Learned Perceptual Image Patch Similarity (Zhang et al., 2018) (LPIPS), used to evaluate consistency with human visual perception (lower scores indicate better performance). **No-reference metrics:** Blind/Referenceless Image Spatial Quality Evaluator (Mittal et al., 2012) (BRISQUE) and Natural Image Quality Evaluator (Mittal et al., 2013) (NIQE), used to assess the absolute visual quality of generated images (lower scores indicate better performance).

4.1.3. Comparison methods

To demonstrate the superiority of our approach, we compare MCLE-Net with a wide range of existing LLIE algorithms, covering mainstream technical routes including Retinex-based, Transformer-based, and diffusion-based models. The compared methods include SCI (Ma et al., 2022), RetinexFormer (Cai et al., 2023b), RQ-LLIE (Liu et al., 2023), CUE (Zheng et al., 2023a), PairLIE (Fu et al., 2023), FourLLIE (Wang et al., 2023a), LLFormer (Wang et al., 2023b), DiffLL (Jiang et al., 2023), LightenDiff (Jiang et al., 2024b), and Diff-Retinex++ (Yi et al., 2025). To ensure fairness, all comparison methods use official source code and pre-trained models.

4.1.4. Implementation details

Our model is implemented using Python 3.9 and PyTorch. For training, we use Adam optimizer and a cosine annealing learning rate scheduler with restarts, setting the initial learning rate to 2×10^{-4}. We train the model for 400,000 iterations using a batch size of 16 and a patch size of 256×256. To prevent overfitting and improve model generalization, we apply comprehensive data augmentation strategies. In addition to standard geometric augmentations (e.g., random flipping and rotation), we specifically incorporate mixup augmentation. All experiments are performed on a platform equipped with an Intel Core i9-12900K CPU and an NVIDIA GeForce RTX 4090 GPU. To ensure a strictly fair comparison, all compared methods are evaluated under the exact same conditions. Both our proposed MCLE-Net and all baseline models are evaluated using the identical dataset splits detailed in Table 1. For the compared methods, we use their official source codes and pre-trained weights corresponding to these exact datasets. Furthermore, all quantitative metrics are computed using standard evaluation scripts on the luminance Y channel of the YCbCr color space. For standardized efficiency evaluation, the average inference time (AIT) is measured at a

Table 2
Quantitative comparison with other methods on the LoLI-Street-Light dataset, with optimal and suboptimal results highlighted in underline and **bold**.

LoLI-Street-Light						
Methods	PSNR↑ (dB)	SSIM↑	MS-SSIM↑	LPIPS↓	MSE↓	MAE↓
SCI CVPR'22	27.84	0.8759	0.9413	0.0429	106.87	206.12
RetinexF. ICCV'23	27.92	0.8767	0.9422	0.0419	106.79	206.09
RQ-LLIE ICCV'23	27.66	0.8811	0.9035	0.1038	112.01	**55.35**
CUE ICCV'23	28.77	0.6493	0.3177	0.4194	89.34	125.41
PairLIE CVPR'23	28.78	0.9169	0.9372	0.0625	86.62	84.86
FourLLIE ACM'23	28.06	0.8363	0.8196	0.1996	101.99	87.12
LLFormer AAAI'23	**33.40**	0.9648	0.9876	0.0039	**30.29**	93.59
DiffLL TOG'23	32.59	0.9560	**0.9889**	**0.0139**	35.82	115.68
LightenDiff ECCV'2024	32.81	0.9577	0.9824	0.0247	34.05	67.23
Diff-Retinex++ TPAMI'25	33.37	0.9581	0.9871	0.0214	30.72	64.82
MCLE-Net	34.08	**0.9635**	0.9922	0.0191	29.01	44.04

Table 3
Quantitative comparison with other methods on the LoLI-Street-Moderate dataset, with optimal and suboptimal results highlighted in underline and **bold**.

LoLI-Street-Moderate						
Methods	PSNR↑ (dB)	SSIM↑	MS-SSIM↑	LPIPS↓	MSE↓	MAE↓
SCI CVPR'22	28.38	0.6192	0.7809	0.1572	94.88	169.69
RetinexF. ICCV'23	28.44	0.6197	0.7819	0.1563	94.78	169.68
RQ-LLIE ICCV'23	27.59	0.8829	0.9372	0.0668	113.49	44.33
CUE ICCV'23	30.58	0.9100	0.9668	0.0201	57.41	148.72
PairLIE CVPR'23	28.42	0.9242	0.9374	0.0447	93.75	162.81
FourLLIE ACM'23	27.96	0.8693	0.8336	0.1831	104.14	113.81
LLFormer AAAI'23	32.15	0.9386	0.9837	0.0061	40.14	80.25
DiffLL TOG'23	31.87	0.9352	0.9789	**0.0142**	42.27	138.32
LightenDiff ECCV'2024	32.09	0.9342	0.9661	0.0397	40.19	58.27
Diff-Retinex++ TPAMI'25	**32.57**	**0.9388**	**0.9839**	0.0331	**34.79**	**49.21**
MCLE-Net	33.61	0.9479	0.9885	0.0297	31.32	50.37

fixed 400 × 600 resolution and a batch size of 1. To isolate pure architectural efficiency, timing captures only the GPU forward pass. All evaluations are performed in FP32 precision.

4.2. Quantitative and qualitative analysis: LoLI-street dataset

In this section, we conduct a comprehensive quantitative and qualitative evaluation of MCLE-Net and existing methods on the LoLI-Street dataset.

4.2.1. Quantitative results

We conduct experiments on three subsets of LoLI-Street (Light, Moderate, Dense), with quantitative results presented in Tables 2–4.

As shown in Table 2, on the LoLI-Street-Light dataset, MCLE-Net yields highly competitive results across multiple key metrics. Notably, it achieves the best scores in PSNR, MS-SSIM, MSE, and MAE, validating the effectiveness of the model in restoring both structural fidelity and pixel-level accuracy. Meanwhile, its SSIM and LPIPS scores remain highly comparable to the top-performing baselines, demonstrating a well-balanced performance.

On the more challenging LoLI-Street-Moderate dataset, as shown in Table 3, the images contain complex shadows and reduced brightness, which places higher demands on enhancement algorithms. MCLE-Net consistently delivers strong performance, securing the top rank across four metrics: PSNR, SSIM, MS-SSIM, and MSE. In particular, it attains a PSNR of 33.61 dB, providing a margin over the next-best method and confirming its robustness in complex scenes.

As shown in Table 4, the most challenging LoLI-Street-Dense dataset features extremely low image brightness and contains numerous densely packed objects. Under these severe conditions, MCLE-Net exhibits notable robustness, achieving the leading results across four key metrics.

Table 4
Quantitative comparison with other methods on the LoLI-Street-Dense dataset, with optimal and suboptimal results highlighted in underline and **bold**.

LoLI-Street-Dense						
Methods	PSNR↑ (dB)	SSIM↑	MS-SSIM↑	LPIPS↓	MSE↓	MAE↓
SCI CVPR'22	27.79	0.7398	0.8621	0.3037	108.02	142.15
RetinexF. ICCV'23	27.87	0.8767	0.9422	0.0419	106.79	206.09
RQ-LLIE ICCV'23	29.03	0.9167	0.9616	0.0326	82.51	107.57
CUE ICCV'23	30.58	0.9100	0.9668	**0.0201**	57.41	148.72
PairLIE CVPR'23	27.66	0.8702	0.9413	0.0357	111.57	214.17
FourLLIE ACM'23	28.07	0.8828	0.8780	0.1188	101.52	186.56
LLFormer AAAI'23	31.62	0.9274	0.9738	0.0131	45.39	108.04
DiffLL TOG'23	31.04	0.9165	**0.9734**	0.0273	51.66	123.79
LightenDiff ECCV'2024	31.89	0.9112	0.9605	0.0327	42.08	88.07
Diff-Retinex++ TPAMI'25	**32.06**	0.9177	0.9692	0.0311	**40.47**	**84.26**
MCLE-Net	33.40	**0.9202**	0.9813	0.0299	30.22	80.10

Table 5
Quantitative comparison with other methods on the LoLI-Street real low-light test set, with optimal and suboptimal results highlighted in underline and **bold**.

Metrics	Pre-trained weights		Trained weights	
Methods	BRISQUE↓	NIQE↓	BRISQUE↓	NIQE↓
SCI CVPR'22	38.05	12.16	30.96	12.32
RetinexF. ICCV'23	54.12	11.79	41.69	11.83
RQ-LLIE ICCV'23	29.76	12.57	29.76	12.57
CUE ICCV'23	13.65	14.36	25.97	12.44
PairLIE CVPR'23	39.65	12.28	35.25	**11.78**
FourLLIE ACM'23	15.05	12.57	**14.50**	11.89
LLFormer AAAI'23	**12.69**	16.40	30.44	11.84
DiffLL TOG'23	18.54	12.16	30.11	12.30
LightenDiff ECCV'2024	13.77	12.09	15.57	12.04
Diff-Retinex++ TPAMI'25	12.09	**11.88**	14.91	11.94
MCLE-Net	–	–	13.85	11.01

This strong performance is primarily attributed to the proposed MDAB module. Its coordinated processing across the spatial, channel, and frequency domains effectively resolves the severe degradation typically encountered in high-density scenes. Furthermore, LLFormer achieves the best LPIPS score. This likely stems from the global receptive field of the Transformer architecture, which grants it an advantage in generating images with enhanced perceptual smoothness.

No reference image low light enhancement results: To evaluate the generalization capability of the model in real-world scenarios, we conduct evaluations on the LoLI-Street unpaired real low-light test set. The results are presented in Table 5.

When using official pre-trained weights, LLFormer and Diff-Retinex++ deliver strong performance on both the BRISQUE and NIQE metrics, showcasing their robust foundation as general-purpose models. However, after fine-tuning on the LoLI-Street dataset, MCLE-Net achieves the best NIQE score, yielding highly competitive results against the baselines. This indicates that MCLE-Net generates images with enhanced naturalness and reduced perceptual distortion, which is crucial for demanding applications such as autonomous driving.

Furthermore, after domain-specific training, the previous leading methods (RetinexFormer and Diff-Retinex++) attain NIQE scores of 11.83 and 11.94, respectively, both of which are higher (worse) than that of MCLE-Net. This demonstrates that our method can more effectively leverage scenario-specific data for deep optimization and better adapt to the image characteristics of the target domain.

4.2.2. Qualitative results

To qualitatively assess MCLE-Net, we compare it with existing models on the Light, Moderate, and Dense subsets of LoLI-Street. As shown in Figs. 5–7, our method effectively balances detail restoration, noise suppression, and color fidelity.

Fig. 5. Qualitative comparison with other methods on the LoLI-street-light dataset.

First, MCLE-Net demonstrates a strong capability to recover fine details and sharp edges. While competing methods often introduce blurriness or granular artifacts when increasing brightness, and even baselines like Diff-Retinex++ struggle with distant building details (Fig. 5), our approach explicitly preserves high-frequency information. Specifically, as observed in the zoomed-in regions of Figs. 5–7, key elements such as building textures, lane markings, and license plate text are sharper. This precise texture recovery is directly attributed to the frequency-domain processing within our MDAB module, which effectively compensates for high-frequency attenuation without inadvertently amplifying local noise.

Secondly, MCLE-Net exhibits noise suppression in severely degraded scenes. As shown in Fig. 7, when attempting to restore clarity to objects such as vehicles and traffic signs, several baselines inevitably introduce noise or unnatural artifacts. In contrast, our approach effectively avoids noise amplification, thereby maintaining image purity. This capability stems from our adaptive Top-k sparse attention mechanism, which forces the network to focus on salient structural correlations and discard redundant, noisy features.

Furthermore, visual assessments indicate that MCLE-Net maintains consistent color fidelity and illumination contrast. Several competing methods suffer from noticeable color shifts (e.g., SCI) or produce grayish images with insufficient contrast (e.g., RetinexFormer, LLFormer). Unlike these approaches, our model brightens shadowed areas while accurately restoring natural colors, effectively avoiding color distortion or oversaturation (Fig. 6). This balanced illumination adjustment is facilitated by the global channel semantics modeling in our framework, which comprehensively captures holistic color distributions.

Finally, these visual comparisons confirm that MCLE-Net provides robust adaptability across diverse and challenging scenarios. By collaboratively processing information across spatial, channel, and frequency domains, our model effectively handles varying low-light conditions, ranging from light to severe, delivering stable visual outputs and confirming its excellent generalization capability.

4.3. Quantitative and qualitative analysis: general datasets

In this section, to further validate the generalization capability and robustness of MCLE-Net, we conduct comprehensive quantitative and qualitative evaluations on multiple widely used LLIE benchmark datasets, including LOL v1, LOL v2 (real and synthetic), and LSRW.

4.3.1. Quantitative results

As shown in Tables 6–9, MCLE-Net delivers competitive performance across four datasets, consistently achieving top-tier results on most key metrics.

On the LOL v1 dataset (Table 6), our method achieves the best results on four metrics: PSNR, MS-SSIM, LPIPS, and MSE. The improvement in PSNR is particularly notable, as our model surpasses the next-best method, Diff-Retinex, by a margin of 1.1 dB, indicating a clear enhancement in reconstruction fidelity.

While CUE records the highest SSIM score and SCI achieves the lowest MAE, the performance of the proposed algorithm remains exceptionally competitive in these areas, showing strong advantages in metrics that measure pixel-level accuracy (PSNR, MSE) and perceptual quality (LPIPS). The leading LPIPS score, in particular, suggests that the images enhanced by our model are perceptually more pleasing and closer to the GT. This consistent and robust performance on the LOL v1 dataset firmly establishes the effectiveness and generalization ability of our proposed approach.

On the challenging LOL v2-real dataset (Table 7), MCLE-Net demonstrates leading performance by surpassing competing methods across the vast majority of metrics. Notably, it achieves a PSNR of 29.87 dB and an SSIM of 0.8909, exceeding the recent strong performer Diff-Retinex by a margin of 0.7 dB in PSNR. This consistent advantage across metrics that measure fidelity, structural integrity, and perceptual quality highlights the exceptional ability of our model to recover details and structure in real-world low-light conditions.

The results on the LOL v2-synthetic dataset, summarized in Table 8, further underscore the robustness and effectiveness of our approach.

Fig. 6. Qualitative comparison with other methods on the LoLI-street-moderate dataset.

Fig. 7. Qualitative comparison with other methods on the LoLI-street-dense dataset.

MCLE-Net again achieves leading performance on the majority of the metrics, securing the top scores for PSNR (29.89 dB), MS-SSIM (0.9738), LPIPS (0.0401), and MSE (71.52). It is noteworthy that while FourLLIE obtains the best results for SSIM and MAE, the performance of our method in PSNR and, crucially, in LPIPS, indicates a better balance between pixel-level accuracy and perceptual realism. These quantitative results demonstrate the effectiveness of MCLE-Net for LLIE tasks.

On the large-scale real-world LSRW dataset (Table 9), our proposed MCLE-Net continues to exhibit competitive performance, securing the best results on a majority of the key metrics. Specifically, our method achieves a PSNR of 29.04 dB, an MS-SSIM of 0.7907, and the lowest (best) error scores with an MSE of 82.33 and an MAE of 168.48. This demonstrates the robust capability of our model in achieving high-fidelity image reconstruction and minimizing pixel-level errors, even when generalizing to a new data domain.

While FourLLIE and PairLIE achieve the leading scores in the SSIM and LPIPS metrics, respectively, the performance of our method in MS-SSIM, a metric often considered more advanced and perceptually

Table 6
Quantitative comparison with other methods on the LOL v1 dataset, with optimal and suboptimal results highlighted in underline and **bold**.

LOL v1						
Methods	PSNR↑(dB)	SSIM↑	MS-SSIM↑	LPIPS↓	MSE↓	MAE↓
SCI CVPR'22	27.95	0.6333	0.7456	0.2177	104.42	150.58
RetinexF. ICCV'23	27.89	0.6299	0.7542	0.2072	106.44	174.08
RQ-LLIE ICCV'23	27.97	0.8724	0.8562	0.1418	104.59	176.54
CUE ICCV'23	27.82	0.7111	0.8313	0.1448	108.18	179.70
PairLIE CVPR'23	27.93	0.7074	0.7932	0.1595	105.25	182.58
FourLLIE ACM'23	28.00	0.8181	0.8642	0.1157	105.14	172.88
LLFormer AAAI'23	27.77	0.7778	0.8389	0.1502	109.08	179.50
DiffLL TOG'23	27.88	0.8207	0.8555	0.1473	105.02	169.74
LightenDiff ECCV'2024	27.92	0.7978	0.8581	0.1402	104.08	174.50
Diff-Retinex++ TPAMI'25	**28.12**	0.8278	**0.8817**	**0.1108**	**99.74**	160.78
MCLE-Net	29.22	**0.8596**	0.9468	0.0832	80.26	**158.66**

Table 7
Quantitative comparison with other methods on the LOL v2-real dataset, with optimal and suboptimal results highlighted in underline and **bold**.

LOL v2-real						
Methods	PSNR↑(dB)	SSIM↑	MS-SSIM↑	LPIPS↓	MSE↓	MAE↓
SCI CVPR'22	27.73	0.7543	0.8242	0.3518	109.94	176.08
RetinexF. ICCV'23	27.82	0.7650	0.8261	0.5583	107.39	170.48
RQ-LLIE ICCV'23	27.82	0.7799	0.8753	**0.0988**	109.26	180.82
CUE ICCV'23	28.13	0.8437	0.8951	0.1057	102.06	174.28
PairLIE CVPR'23	27.77	0.7246	0.8612	0.1021	109.78	199.54
FourLLIE ACM'23	27.62	0.7473	0.8167	0.1580	112.71	205.84
LLFormer AAAI'23	27.72	0.7670	0.8651	0.1163	110.77	185.34
DiffLL TOG'23	27.88	0.7875	0.8695	0.1219	108.32	175.12
LightenDiff ECCV'2024	28.41	0.8271	0.8733	0.1093	94.73	178.34
Diff-Retinex++ TPAMI'25	**29.17**	**0.8426**	**0.9031**	0.1073	**78.03**	157.09
MCLE-Net	29.87	0.8909	0.9621	0.0750	69.19	**169.34**

Table 8
Quantitative comparison with other methods on the LOL v2-synthetic dataset, with optimal and suboptimal results highlighted in underline and **bold**.

LOL v2-synthetic						
Methods	PSNR↑(dB)	SSIM↑	MS-SSIM↑	LPIPS↓	MSE↓	MAE↓
SCI CVPR'22	28.03	0.6557	0.7059	0.2969	103.17	176.12
RetinexF. ICCV'23	28.02	0.6511	0.7756	0.2449	103.06	158.50
RQ-LLIE ICCV'23	28.07	0.6392	0.7741	0.2470	102.03	164.45
CUE ICCV'23	28.01	0.6516	0.7905	0.2397	103.26	159.11
PairLIE CVPR'23	28.14	0.8223	0.8623	0.1697	101.06	167.85
FourLLIE ACM'23	**29.61**	0.9623	**0.9656**	**0.0403**	**76.78**	121.24
LLFormer AAAI'23	28.03	0.6759	0.7799	0.2374	103.01	156.94
DiffLL TOG'23	28.03	0.5967	0.7644	0.2567	102.85	159.62
LightenDiff ECCV'2024	28.69	0.8559	0.8899	0.2074	90.25	151.55
Diff-Retinex++ TPAMI'25	29.07	0.8971	0.9014	0.1203	80.01	151.50
MCLE-Net	29.89	**0.9491**	0.9738	0.0401	71.52	**148.86**

relevant than single-scale SSIM, suggests a stronger capability in preserving structural integrity across multiple resolutions. In summary, the strong overall performance on the LSRW dataset underscores the broad generalization ability of our proposed MCLE-Net, confirming its effectiveness and practical utility beyond standard benchmark datasets.

4.3.2. Qualitative results

For a direct visual comparison, Figs. 8 and 9 present the enhancement results of MCLE-Net against various existing methods on the four public datasets.

Fig. 8 shows qualitative comparisons on the LOL v1 and LSRW datasets. Visual assessments indicate that the results generated by our method are highly consistent with the GT, showing an effective balance

Table 9
Quantitative comparison with other methods on the LSRW dataset, with optimal and suboptimal results highlighted in underline and **bold**.

LSRW						
Methods	PSNR↑(dB)	SSIM↑	MS-SSIM↑	LPIPS↓	MSE↓	MAE↓
SCI CVPR'22	27.81	0.5938	0.6399	0.3526	107.88	172.28
RetinexF. ICCV'23	27.96	0.6160	0.6832	0.1873	104.56	182.63
RQ-LLIE ICCV'23	28.00	0.6590	0.6900	0.1648	103.52	176.41
CUE ICCV'23	27.93	**0.6751**	0.6959	**0.1592**	102.69	177.46
PairLIE CVPR'23	27.94	0.6684	0.7045	0.1358	105.12	185.33
FourLLIE ACM'23	27.98	0.7298	0.6970	0.1748	103.77	174.83
LLFormer AAAI'23	27.95	0.6543	0.6958	0.1685	104.77	182.88
DiffLL TOG'23	27.98	0.6099	0.6846	0.1805	104.10	172.70
LightenDiff ECCV'2024	28.77	0.6443	0.6858	0.1789	88.34	171.88
Diff-Retinex++ TPAMI'25	**28.91**	0.6503	**0.7107**	0.1689	**84.69**	**169.23**
MCLE-Net	29.04	0.6576	0.7907	0.1642	82.33	168.48

of brightness, contrast, and color fidelity. For instance, on the LSRW dataset, it accurately restores the white body of the motorcycle and the ambient colors while appropriately boosting overall brightness. In contrast, methods like SCI and RetinexFormer exhibit insufficient brightness or color deviation. At the detail level, our method recovers sharp and well-defined textures, effectively suppressing blur and artifacts, which is particularly evident when compared to LightenDiff and Diff-Retinex++.

The results on the LOL v2 dataset, shown in Fig. 9, further validate the effectiveness of MCLE-Net. Its advantages are particularly prominent in detail recovery. In text restoration cases, our approach generates highly legible text structures that closely align with the GT, avoiding the severe blurring observed in methods like RQ-LLIE and LightenDiff. It also preserves complex building textures and sculptural details with high fidelity. These results indicate the robustness of our model in restoring text information and texture details, which are crucial for high-level vision tasks. This precise high-frequency recovery is directly driven by the frequency-domain processing within the MDAB module. Furthermore, MCLE-Net performs robustly in color and brightness rendition. It accurately restores natural hues and proper exposure to the scenes, thereby effectively avoiding common issues like color casts, insufficient saturation, or imbalanced contrast found in other approaches. This stable color reproduction further highlights the efficacy of the global channel semantics modeling in maintaining holistic color distributions.

4.4. Ablation study

4.4.1. Ablation studies on the proposed components

To validate the effectiveness of key components in the proposed MCLE-Net, we conduct an ablation study on the contributions of MDABs and GFBs, evaluating four model variants: (i) a baseline network without either MDAB or GFB, (ii) the baseline with only GFB, (iii) the baseline with only MDAB, and (iv) the full MCLE-Net architecture with both modules.

Quantitative results: As summarized in Table 10, the baseline model (w/o MDAB, w/o GFB) achieves a PSNR of 23.72 dB. Integrating either the GFB (+2.61 dB) or the MDAB (+1.67 dB) alone yields significant performance gains. The full model, incorporating both components, achieves the best performance across all metrics (e.g., 29.22 dB PSNR). This demonstrates a powerful synergistic effect between MDAB and GFB, as their combined improvement surpasses the sum of their individual contributions.

Qualitative results: The visual results as shown in Fig. 10. The baseline model suffers from significant detail loss and noise. While individual components improve illumination (GFB) or detail recovery (MDAB), they fail to achieve a comprehensive enhancement. In contrast, the complete MCLE-Net model successfully restores the scene with natural colors, balanced contrast, and high-fidelity details, rendering the text on the sign sharp and legible, closely approaching the GT.

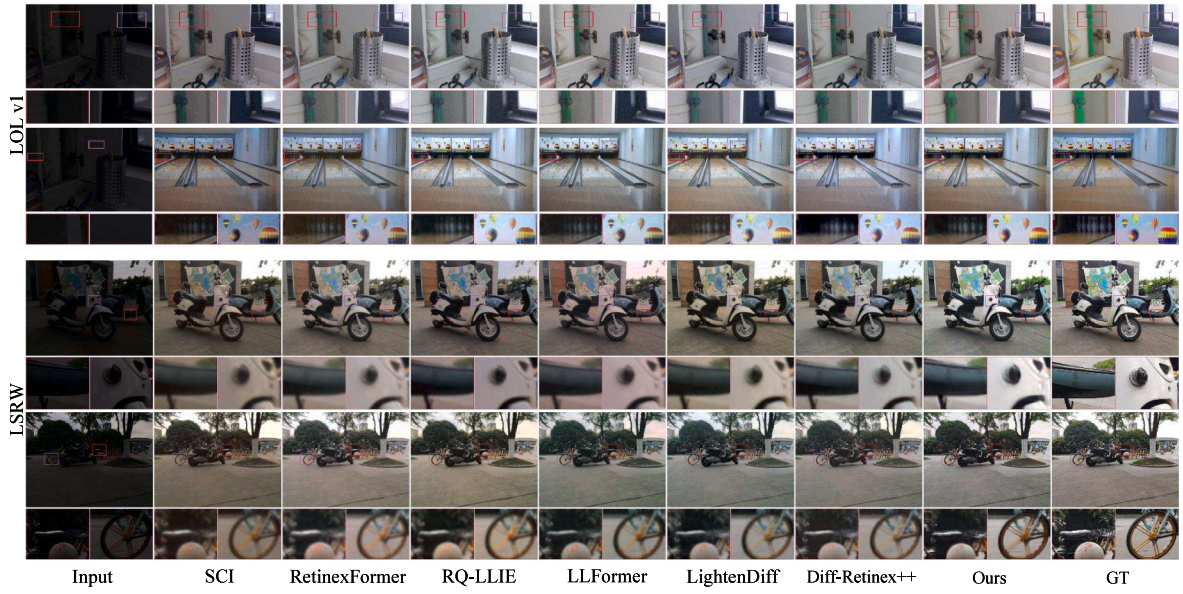

Fig. 8. Qualitative comparison with other methods on the LOL v1 and LSRW dataset.

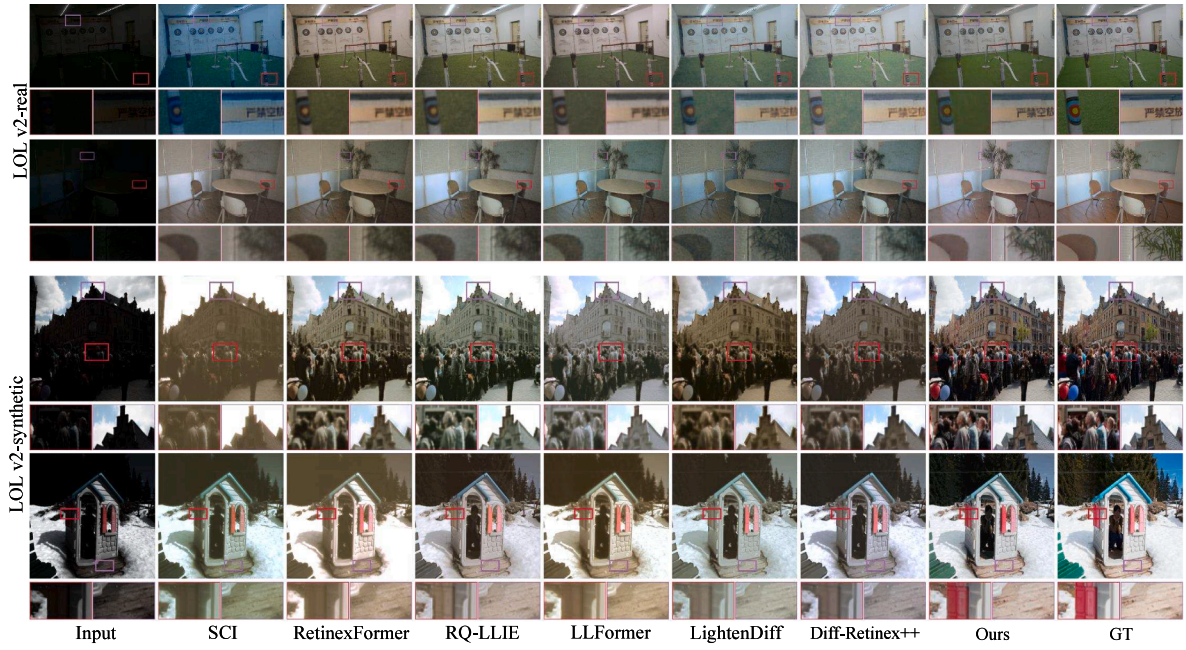

Fig. 9. Qualitative comparison with other methods on the LOL v2 dataset.

Table 10
Ablation study of the MCLE-Net module, with optimal and suboptimal results highlighted in underline and **bold**.

Methods	PSNR↑(dB)	MS-SSIM↑	LPIPS↓	MAE↓	Params↓	FLOPS↓	AIT↓
w/o MDAB, *w/o* GFB	23.72	0.8879	0.1364	176.91	12.83M	40.12G	0.077s
w MDAB, *w/o* GFB	**26.33**	**0.9172**	**0.1092**	**170.11**	25.78M	52.77G	**0.109s**
w/o MDAB, *w* GFB	25.39	0.9031	0.1107	171.39	**17.63M**	**49.36G**	0.114s
w MDAB, *w* GFB(Ours)	29.22	0.9468	0.0832	168.66	34.59M	66.78G	0.169s

4.4.2. *Ablation on sub-components of MDAB*

Table 11 presents an ablation study evaluating the individual contributions of the sub-components within the MDAB. The baseline model excludes the SCA, SSA, and the frequency branch. Introducing the SCA efficiently improves the recovery of color and illumination distributions with minimal computational overhead. Further integrating the SSA leads to substantial performance gains. Although this step introduces additional computational complexity, it proves essential for capturing long-range spatial dependencies and restoring fine structural details. Finally, incorporating the frequency branch to complete the MDAB

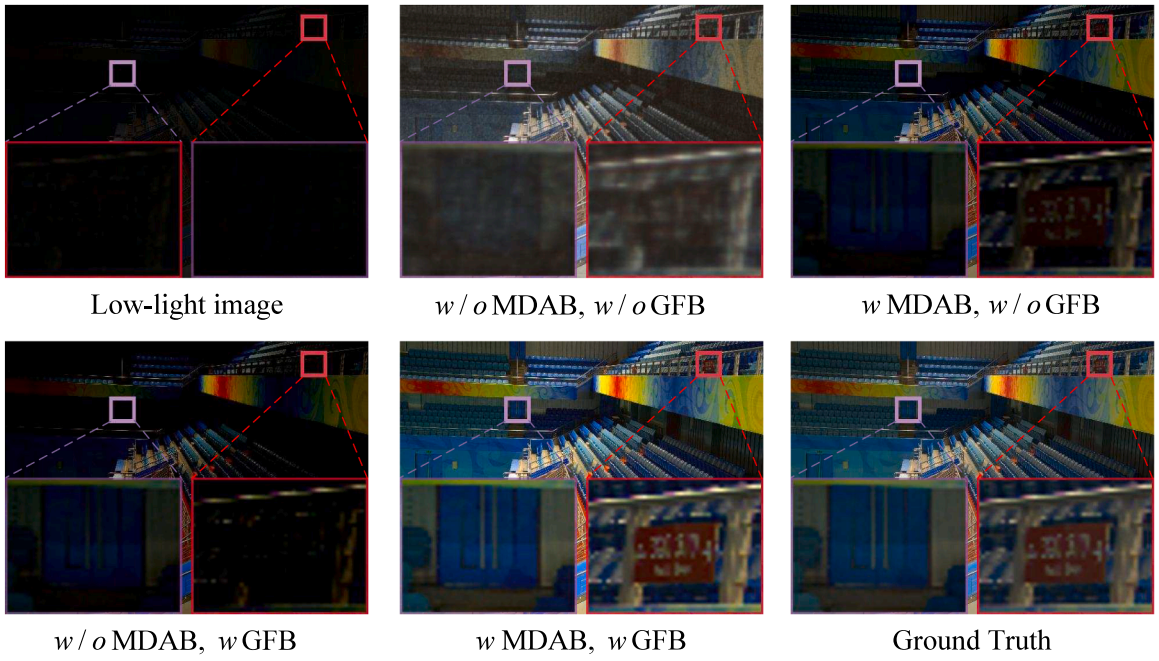

Fig. 10. Visualization of ablation study results.

Table 11
Detailed ablation study on the sub-components of MDAB.

SCA	SSA	Freq Branch	PSNR↑(dB)	MS-SSIM↑	LPIPS↓	MAE↓	Params↓	FLOPS↓	AIT↓
			25.39	0.9031	0.1107	171.39	17.63M	49.36G	0.114s
✓			26.21	0.9145	0.1035	170.52	19.88M	52.86G	0.126s
✓	✓		28.36	0.9352	0.0915	169.45	31.65M	62.36G	0.164s
✓	✓	✓	29.22	0.9468	0.0832	168.66	34.59M	66.78G	0.169s

architecture achieves the optimal overall performance. By explicitly processing frequency components, this branch further refines global illumination and preserves structural fidelity, leading to minimized perceptual and pixel-level errors. These progressive improvements demonstrate that the synergistic combination of spatial, channel, and frequency-domain modeling forms an effective and comprehensive architecture for low-light image restoration.

4.4.3. Ablation on frequency-domain processing strategies and loss

Table 12 presents an ablation study evaluating different frequency-domain processing strategies and the impact of the frequency loss. Building upon the baseline of spatial and channel domains ("w/o Freq Branch"), incorporating frequency-domain operations requires specific architectural constraints. Specifically, allowing the network to modify the phase spectrum ("w Phase, w/o Amp") leads to a significant performance degradation, indicating that altering original phase information disrupts inherent spatial structures. Even when both amplitude and phase are modified simultaneously ("w Phase, w Amp"), the model still underperforms the baseline, further confirming the negative impact of phase distortion. In contrast, the proposed method ("w/o Phase, w Amp (Ours)"), which preserves the phase and corrects only the amplitude, achieves the optimal performance. Furthermore, removing the frequency-domain loss ("w/o Freq Loss") results in suboptimal outcomes, demonstrating that explicit frequency supervision is necessary for guiding the amplitude correction process. These comparisons indicate that the phase-preserving amplitude adjustment strategy is the key to maintaining structural fidelity during enhancement.

4.4.4. Analysis of architectural complexity

Table 13 presents an analysis of architectural complexity by varying the multi-scale depth and the amplitude subnetwork capacity. Regarding the multi-scale structure, a shallower network (Depth = 2) restricts the receptive field and limits hierarchical feature abstraction, leading to insufficient restoration capabilities. Conversely, expanding the depth (Depth = 4) yields higher quantitative metrics but results in a disproportionate surge in parameter count, indicating diminishing returns in computational efficiency. Similarly, modulating the amplitude subnetwork reveals that a reduced channel width (×0.5) limits the representational capacity necessary for modeling complex frequency-domain amplitude corrections. While doubling the channels (×2.0) provides further performance gains, it introduces additional architectural overhead. These structural comparisons demonstrate that the proposed configuration establishes an effective trade-off between computational cost and image enhancement quality.

Table 12
Ablation study on different frequency-domain processing strategies and the frequency loss.

Method	PSNR↑(dB)	MS-SSIM↑	LPIPS↓	MAE↓
w/o Freq Branch	28.36	0.9352	0.0915	169.45
w Phase, w/o Amp	25.50	0.9035	0.1152	175.40
w Phase, w Amp	27.85	0.9155	0.0985	171.25
w/o Freq Loss	28.95	0.9395	0.0875	169.05
w/o Phase, w Amp (Ours)	29.22	0.9468	0.0832	168.66

4.4.5. Ablation on the choice of the threshold (τ)

To evaluate how sensitive the model performance is to the choice of this threshold, which determines the effective k, we conduct an ablation study by varying τ from 0.4 to 0.8. The results on the LOL v1 dataset are presented below.

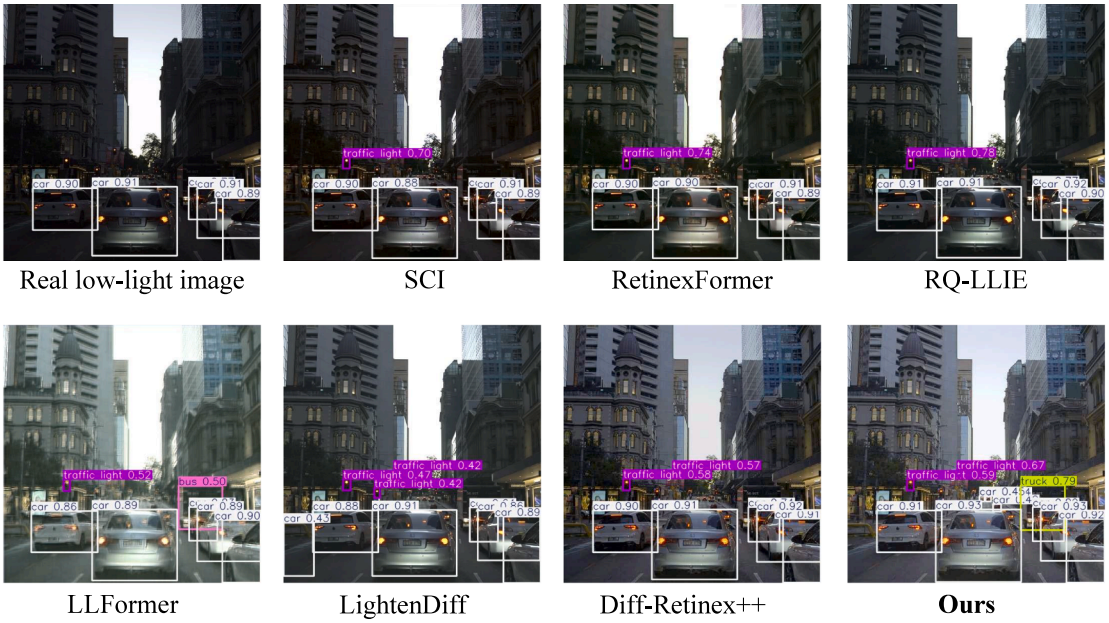

Fig. 11. Qualitative comparison of the impact of various image enhancement algorithms on downstream object detection performance in real-world low-light scenarios.

Table 13
Analysis of architectural complexity by varying the multi-scale depth and amplitude subnetwork capacity. The optimal result is highlighted in bold.

	Configuration	PSNR↑(dB)	MS-SSIM↑	Params↓
Multi-scale Depth	Depth = 2	27.94	0.9312	21.35M
	Depth = 3 (Ours)	**29.22**	**0.9468**	**34.59M**
	Depth = 4	30.57	0.9580	58.42M
Amplitude Capacity	Subnetwork Channels ×0.5	28.77	0.9415	33.27M
	Subnetwork Channels ×2.0	29.63	0.9472	36.84M
	Ours	**29.22**	**0.9468**	**34.59M**

Table 14
Ablation study on the choice of the threshold (τ).The optimal result is highlighted in bold.

τ	PSNR↑(dB)	MS-SSIM↑	LPIPS↓	MAE↓	FLOPS↓	AIT↓
0.4	28.65	0.9315	0.0982	174.85	68.28G	0.181s
0.5	29.05	0.9412	0.0875	170.12	67.53G	0.176s
0.6	**29.22**	**0.9468**	**0.0832**	**168.66**	**66.78G**	**0.169s**
0.7	28.93	0.9388	0.0915	172.05	65.81G	0.164s
0.8	28.51	0.9205	0.1085	175.62	64.95G	0.163s

As shown in Table 14, the model is sensitive to the selection of this threshold. When τ is set too low (e.g., 0.4), the attention map fails to achieve sufficient sparsity, retaining redundant local noise amplified during enhancement. Conversely, setting τ too high (e.g., 0.8) overly sparsifies the matrix, discarding critical structural correlations and leading to a performance drop. The threshold of 0.6 provides the optimal balance between computational efficiency and high-fidelity texture preservation.

4.4.6. Sensitivity analysis of loss weights

Table 15 presents a sensitivity analysis of the hyperparameters λ_2 and λ_3 in the hybrid loss function, employing a one-factor-at-a-time strategy with the primary spatial loss weight λ_1 fixed to 1.0. The top half of the table illustrates the impact of varying the frequency loss weight λ_2 while keeping λ_3 at 0.01. Since the unnormalized FFT spectrum inherently produces large gradient magnitudes, setting λ_2 to an excessive value (e.g., 0.5) overpowers the spatial optimization and disrupts the overall image reconstruction, causing a decrease in PSNR. Conversely,

Table 15
Sensitivity analysis of the loss weights λ_2 and λ_3 with λ_1 fixed to 1.0. The optimal configuration is highlighted in bold.

Setting	λ_1	λ_2	λ_3	PSNR	MS-SSIM
Varying λ_2					
	1	0.01	0.01	28.88	0.9405
	1	0.05	0.01	29.02	0.9435
	1	**0.1**	0.01	**29.22**	**0.9468**
	1	0.2	0.01	28.87	0.9380
	1	0.5	0.01	28.24	0.9215
Varying λ_3					
	1	0.1	0.001	29.01	0.9425
	1	0.1	0.005	29.15	0.9442
	1	0.1	**0.01**	**29.22**	**0.9468**
	1	0.1	0.02	29.06	0.9440
	1	0.1	0.05	28.89	0.9432

setting λ_2 too small (e.g., 0.01) weakens the frequency-domain supervision, leading to a suboptimal PSNR. The bottom half of the table displays the results of varying the structural loss weight λ_3 with λ_2 fixed to 0.1. A larger structural weight (e.g., 0.05) may restrict the pixel-level color and illumination restoration by potentially conflicting with the primary L_1 loss during the optimization process, yielding a lower PSNR. A smaller value (e.g., 0.001) diminishes the structural regularization effect. These empirical results demonstrate that the proposed configuration ($\lambda_1 = 1.0$, $\lambda_2 = 0.1$, $\lambda_3 = 0.01$) effectively balances the gradients of distinct loss terms and provides optimal restoration performance.

4.5. Application to downstream object detection

A central motivation of our MCLE-Net is to improve visual quality to facilitate downstream perception tasks for autonomous driving safety. To validate the effectiveness of our proposed method in improving the accuracy of high-level vision tasks, we specifically conduct experiments on the LoLI-Street real low-light dataset for the task of object detection. YOLOv8 (Jocher et al., 2023) is selected as the foundational object detection method, and the officially provided pre-trained model is utilized directly.

Table 16
Quantitative comparison of object detection performance using YOLOv8, with optimal and suboptimal results highlighted in underline and **bold**.

Methods	SCI	RetinexF.	RQ-LLIE	CUE	PairLIE	FourLLIE	LLFormer	LightenDiff	Diff-R++	MCLE-Net
mAP@0.50↑	0.744	0.758	0.702	0.786	0.748	0.783	0.723	0.811	**0.832**	0.887
mAP@0.50-0.95↑	0.740	0.722	0.671	0.744	0.685	0.731	0.715	0.789	**0.801**	0.854

Table 17
Performance of MCLE-Net and other methods in computing resource consumption and efficiency, with optimal and suboptimal results highlighted in underline and **bold**.

Methods	SCI	RetinexF.	RQ-LLIE	CUE	PairLIE	FourLLIE	LLFormer	LightenDiff	Diff-R++	MCLE-Net
Params↓	384	1.61M	11.38M	0.25M	0.34M	**0.12M**	24.55M	27.83M	62.80M	34.59M
FLOPS↓	0.14G	15.57G	162.07G	157.3G	89.38G	**1.95G**	39.61G	237.92G	241.14G	66.78G
AIT↓	0.001s	**0.009s**	0.074s	0.104s	1.4304s	0.120s	0.267s	0.481s	1.1601s	0.169s

Quantitative Analysis: As reported in Table 16, feeding unenhanced images into the detector limits the overall accuracy. While existing baselines improve the metrics, our MCLE-Net achieves higher performance gains. Specifically, MCLE-Net obtains an mAP@0.5 of 0.887 and an mAP@0.5-0.95 of 0.854. Compared to the second-best method, Diff-Retinex++, our approach yields a consistent improvement. These quantitative results indicate that MCLE-Net effectively restores critical visual representations.

Qualitative Analysis: Visual comparisons of the detection results, illustrated in Fig. 11, align with the quantitative performance. In the original low-light image, degradation causes the detector to miss distant traffic lights and vehicles. While enhancement baselines (such as SCI and RetinexFormer) partially recover visibility, they show limitations with details, resulting in the detection of only one primary traffic light. Additionally, some methods introduce structural distortions that affect detection accuracy. For example, LLFormer produces a false positive by misclassifying a truck as a "bus".

In comparison, MCLE-Net preserves the structural phase while correcting the amplitude in the frequency domain, which minimizes the introduction of artifacts. Consequently, applying YOLOv8 to our enhanced images maintains confidence scores for nearby cars (e.g., 0.91-0.93) and enables the detection of smaller objects, including several distant traffic lights and a truck (confidence 0.79). These results demonstrate that MCLE-Net is an effective enhancement solution for autonomous driving perception tasks.

4.6. Efficiency analysis

To evaluate the deployment potential of MCLE-Net in practical applications, we compare its model parameters (Params), floating-point operations (FLOPs), and AIT with several existing methods, with the results shown in Table 17.

As can be seen from the table, existing low-light enhancement methods suffer from a clear trade-off between performance and efficiency. On the one hand, while methods like SCI are extremely lightweight, their enhancement performance falls far short of expectations, as demonstrated by the quantitative and qualitative analyses above. On the other hand, some leading methods, such as those based on Transformers and diffusion models, often come with significant computational overhead. In contrast, MCLE-Net strikes an favorable balance between enhancement performance and computational efficiency. Its inference time significantly outperforms other methods with similar performance. In particular, compared to Diff-Retinex++, MCLE-Net is nearly 7 times faster while requiring only a quarter of the computational effort. This result indicates that our method achieves highly competitive performance while effectively controlling computational costs. This excellent trade-off between performance and efficiency highlights the suitability for practical deployment of MCLE-Net in real-world scenarios.

5. Conclusion

To address the issue of image quality degradation in low-light driving scenarios, we propose an efficient and robust MCLE-Net. The core innovation, MDAB, refines image features simultaneously across spatial, channel, and frequency domains. By directly manipulating frequency components via FFT, it achieves precise recovery of high-frequency details and optimization of low-frequency information, effectively resolving common issues in enhancement methods such as blurring, artifacts, and noise amplification. Furthermore, the designed U-shaped MSFEB and GFB ensure effective fusion of multi-scale features and maximize information utilization, enhancing representational capacity of the network and feature integration efficiency.

Through quantitative and qualitative evaluations on multiple synthetic and real-world datasets including LoLI-Street, LOL v1, LOL v2, and LSRW, MCLE-Net demonstrates superior performance across all reference metrics, while improving performance on non-reference metrics. This demonstrates its effective image fidelity, structural similarity, and perceptual quality. Particularly on the LoLI-Street autonomous driving scene dataset, MCLE-Net demonstrates strong detail recovery, noise suppression, and color consistency while maintaining high real-time performance, effectively enhancing the performance of perception systems and driving safety under complex lighting conditions.

Although MCLE-Net has achieved significant progress, future research should explore deeper end-to-end integration optimization with downstream visual tasks (e.g., object detection, semantic segmentation) to enable joint training. Additionally, to address information gaps in extreme low-light conditions, integrating multimodal sensor data (e.g., infrared, lidar) could be explored to enhance model robustness and generalization capabilities.

CRediT authorship contribution statement

Xiaoli Li: Supervision, Funding acquisition; **Zhiyuan Han:** Writing – original draft, Visualization, Validation, Methodology; **Kang Wang:** Writing – review & editing, Supervision, Data curation, Conceptualization; **Guoju Zhang:** Writing – review & editing, Investigation, Data curation; **Zhaohui Liu:** Visualization, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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